Monthly Exam (1) – Model Answers

Question 1

(A) Correct order of the journey of oxygen during breathing:

Nose → trachea → Lungs → (alveoli) → Blood capillaries → Body cells

(B) Choose the correct answer:

1. (b) Oxygen
2. (b) Carbon dioxide
3. (c) Alveoli
4. (c) Small intestine
5. (b) Heart

Put (✓) next to the correct statements and (✗) next to the incorrect statements, correcting the incorrect statements:

1. The inner wall of the small intestine is lined with villi and folds to increase the surface area for absorption. (✓)
2. Carbon dioxide helps combustion to occur. (✗)

Correction: Oxygen supports combustion; carbon dioxide does not support combustion.

3. Air leaves the lungs during inhalation. (✗)

Correction: Air enters the lungs during inhalation.

4. Starch digestion begins in the stomach. (✗)

Correction: Starch digestion begins in the mouth

5. Blood vessels carry blood between the heart and different parts of the body. (✓)
6. Exhaled air contains a greater amount of oxygen than inhaled air. (✗)

Correction: Inhaled air contains more oxygen than exhaled air.

7. Food passes through the esophagus before reaching the stomach. (✓)

Question 3:

Complete the following statements:

A. What is the name of the gas that turns limewater milky?

Answer: Carbon dioxide.

B. What happens to a burning candle flame when the percentage of carbon dioxide in the surrounding air increases?

Answer: The flame becomes weaker and eventually goes out because carbon dioxide does not support burning and the oxygen available is reduced.

C. Blood carries oxygen gas to the body cells.

D. The process of air entering the lungs is called inhalation

E. The process of air leaving the lungs is called exhalation

F. saliva helps food move easily.

G. Food moves from the mouth to the Stomach (through the pharynx and esophagus).

Question 4:

Answer the following:

A. Arrange the following organs in the correct order to show the path of food:

Correct order: Mouth →Pharynx →Esophagus →Stomach →Small intestine → Large intestine.

B. What would happen if the heart did not pump blood to the cells?

The cells would not receive the oxygen and nutrients they need, and wastes would not be carried away. The cells would stop functioning and the body would die.

C. Determine which gas is present in each container (Air – Oxygen gas – Carbon dioxide gas):

Oxygen Air Carbon dioxide

Question 5:

Give reasons for the following:

A. The candle flame goes out when it is covered with a glass beaker.

Because the candle uses up the available oxygen inside the beaker. When there is not enough oxygen to support burning, the flame goes out.

B. When iodine solution is added to the bread, it does not turn blue.

Because the starch in the bread has been broken down by saliva during digestion, so there is no starch left to turn iodine blue-black.

C. We are advised to chew food well.

Chewing breaks food into smaller pieces and mixes it with saliva, making it easier to swallow and helping digestion begin.

Question 6:

Complete the following:

Oxygen gas enters through the nose during inhalation, then passes through the trachea and bronchi until it reaches the lungs, and from there to the alveoli, which are surrounded by blood capillaries. The blood capillaries carry oxygen gas to the body parts, while they carry carbon dioxide gas to the lungs, where it leaves the body during the process of exhalation.